CSA5108 Cryptography and Network security

Name : K. Sirisha
Reg.no:192325130

Exp 18 : DES — Subkey Generation Property
Code:

```
#include <stdio.h>
#include <stdint.h>

int main(void) {
   uint32_t C, D;

   printf("Enter left 28-bit half C in hexadecimal: ");
   scanf("%x",&C);

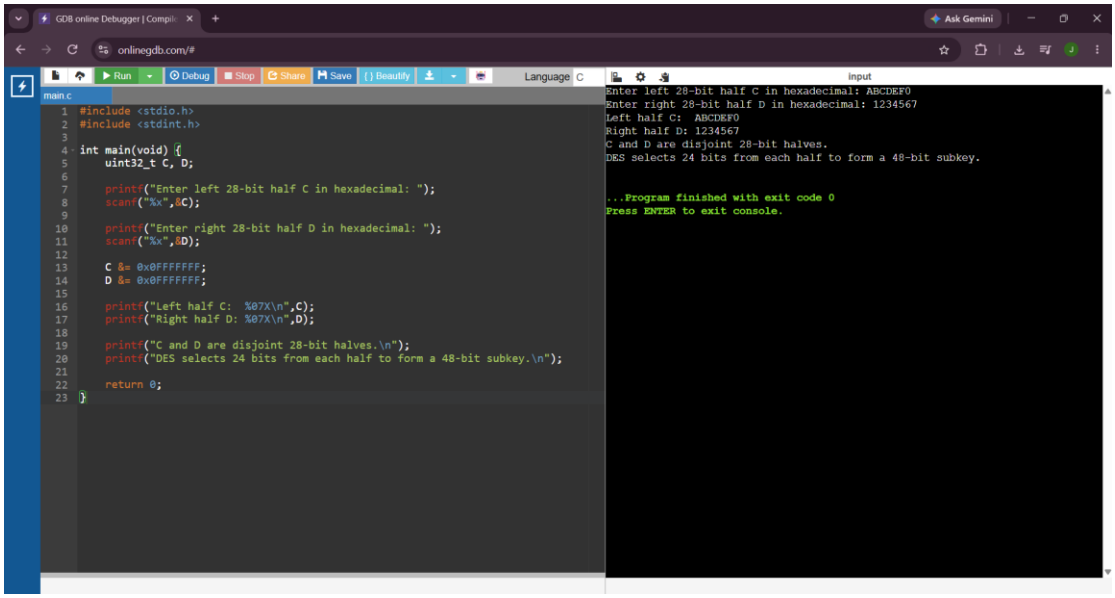
   printf("Enter right 28-bit half D in hexadecimal: ");
   scanf("%x",&D);

   C &= 0x0FFFFFFF;
   D &= 0x0FFFFFFF;

   printf("Left half C:  %07X\n",C);
   printf("Right half D: %07X\n",D);

   printf("C and D are disjoint 28-bit halves.\n");
   printf("DES selects 24 bits from each half to form a 48-bit
subkey.\n");

   return 0;
}
```


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Language C
main.c
#include <stdio.h>
#include <stdint.h>
int main(void) {
uint32_t C, D;
printf("Enter left 28-bit half C in hexadecimal: ");
scanf("%x",&C);
printf("Enter right 28-bit half D in hexadecimal: ");
scanf("%x",&D);
C &= 0x0FFFFFFF;
D &= 0x0FFFFFFF;
printf("Left half C: %07X\n",C);
printf("Right half D: %07X\n",D);
printf("C and D are disjoint 28-bit halves.\n");
printf("DES selects 24 bits from each half to form a 48-bit subkey.\n");
return 0;
}
Input
Enter left 28-bit half C in hexadecimal: ABCDEF0
Enter right 28-bit half D in hexadecimal: 1234567
Left half C: ABCDEF0
Right half D: 1234567
C and D are disjoint 28-bit halves.
DES selects 24 bits from each half to form a 48-bit subkey.
...Program finished with exit code 0
Press ENTER to exit console.